

ORIGINAL ARTICLE

MACB CKD Task Force updated recommendations for reporting estimated glomerular filtration rate and albuminuria in adults

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Abstract

Chronic kidney disease (CKD) is a common clinical condition with significant health risks for patients and is widely recognised as a major public health concern. Laboratory medicine plays a crucial role in both diagnosing and managing CKD, as diagnosis and staging rely on estimated glomerular filtration rate (GFR) and evaluating albuminuria (or proteinuria). It was evident that the laboratory assessment of CKD in Malaysia is not standardised. In light of this, the Malaysian Association of Clinical Biochemistry CKD (MACB-CKD) Task Force issued a national recommendation for laboratory diagnosis of CKD in 2019. Recently, Kidney Disease: Improving Global Outcomes (KDIGO) updated its recommendations for the diagnosis, evaluation, management, and treatment of CKD. These guidelines incorporate the most recent evidence-based practices to support laboratory professionals in delivering optimal care for individuals with CKD, focusing on critical areas such as estimated GFR (eGFR), albuminuria assessment, and risk stratification. The latest National Institute for Health and Care Excellence (NICE) guidelines on CKD has also incorporated the Kidney Failure Risk Equation (KFRE) as a tool for predicting the likelihood of progression to end-stage kidney disease (ESKD) in CKD patients. Hence, the MACB-CKD Task Force has reviewed and updated its recommendations for laboratory reporting of eGFR and urine albumin in alignment with the latest guidelines.

Keywords: CKD, eGFR, albuminuria, KFRE

INTRODUCTION

Chronic kidney disease (CKD) is a major global health issue, especially in low- and middle-income countries, where it is associated with serious health outcomes.¹ Since CKD often advances without noticeable symptoms, clinicians depend on clinical laboratory results to diagnose, classify, treat, and monitor the condition in patients. Laboratory medicine plays a crucial role in diagnosing and managing CKD. Diagnosis and staging are primarily determined through estimated glomerular filtration rate (eGFR) and evaluation of albuminuria.

The Kidney Disease: Improving Global Outcomes (KDIGO) 2012 CKD Clinical Practice Guidelines (CPG)² was built upon the foundational work of the United States-based Kidney Disease Outcomes Quality Initiative (KDOQI) 2002 CPG on the Definition, Classification, and Evaluation of CKD.³ The KDOQI 2002 CPG² was subsequently embraced by the international community in 2005, setting the stage for the development of the more comprehensive KDIGO 2012 CPG. The 2012 CPG introduced the “CGA” classification system for CKD, which categorises the disease based on three factors: cause (C), level of kidney function

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as determined by glomerular filtration rate (GFR) (G), and degree of albuminuria (A). This CGA classification provided a structured approach for the management, treatment, research, and risk assessment of CKD, offering a comprehensive framework that has since become the standard in the field. To address the adherence of Malaysian laboratories to the 2012 KDIGO CPG on the measurement and reporting of eGFR and urine albumin, a national audit was conducted. This audit aimed to assess the current practices and identify any gaps or deviations from the CPG. Given the considerable heterogeneity observed among Malaysian laboratories in the survey results regarding biomarkers of kidney damage, the Malaysian Association of Clinical Biochemists Chronic Kidney Disease (MACB-CKD) Task Force developed recommendations to standardise laboratory reporting of estimated GFR (eGFR) and urine albumin⁴ which were uploaded onto the MACB website on 20 November 2019. These recommendations were aligned with the KDIGO 2012 CPG and the National Institute for Health and Care Excellence (NICE) guidelines.⁵ The release of the KDIGO 2024 CKD CPG⁶ marks a significant update, replacing the 2012 version. This new CPG offers crucial guidance for healthcare professionals on the evaluation and management of CKD and also the key recommendations for clinical laboratories. The MACB-CKD Task Force held a meeting in March 2024 to review and update recommendations on laboratory reporting of eGFR and urine albumin based on KDIGO 2024 and NICE 2022 guidelines. Standardising laboratory reporting is intended to reduce confusion for laboratory users.

Laboratory guidelines for CKD are essential for several reasons:

- (i) Consistency and accuracy: Guidelines ensure that tests are performed and results are reported consistently, improving the accuracy of CKD diagnosis and management across different laboratories.
- (ii) Standardised care: By following established guidelines, laboratories can provide standardised care, ensuring that patients receive uniform diagnostic and monitoring services regardless of location.
- (iii) Early detection and intervention: Standardised testing and reporting enable early detection of CKD, which is crucial for timely intervention and management.
- (iv) Effective monitoring: Guidelines facilitate

effective monitoring of CKD by ensuring that key markers, such as eGFR and albuminuria, are measured and reported consistently.

- (v) Informed decision-making: Accurate and standardised laboratory results help clinicians make informed decisions about treatment plans, adjustments, and management strategies for CKD patients.
- (vi) Quality control: Guidelines help maintain high-quality laboratory practices, reducing the likelihood of errors and ensuring reliable results that support optimal patient care.

By ensuring uniform presentation of results, clinicians can focus on the results rather than questioning discrepancies between laboratories. The advantages of standardisation become evident only with widespread adoption by all laboratories. Achieving this requires collaboration among laboratories. Our main goal was to provide recommendations that can be easily applied in the laboratories in Malaysia. The following sections present recommendations specifically designed for implementation in the clinical laboratory setting.

1. Estimated glomerular filtration rate and serum creatinine

The KDIGO 2024 CKD CPG advises reporting serum creatinine (SCr) levels along with an estimating equation as the initial approach to assess GFR.⁶ The KDIGO 2012 CPG recommended reporting eGFR in adults using the 2009 Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) creatinine equation. However, an alternative creatinine-based GFR estimating equation is considered acceptable if it has been demonstrated to enhance the accuracy of GFR estimates compared to the 2009 CKD-EPI creatinine equation. Estimating equations for GFR have traditionally included demographic variables such as age, sex, and race. These variables are incorporated into the equations to account for variations in serum concentrations of endogenous filtration markers that are not directly related to GFR.

The inclusion of race in eGFR equations, as well as in other algorithms in medicine, has faced growing scrutiny, especially in the United States, but also in other parts of the world. Incorporating race into medical algorithms may perpetuate racial biases and inequalities in healthcare and potentially lead to disparities in diagnosis and treatment. Additionally, there is concern that using race as a variable in

medical calculations may oversimplify complex biological and social determinants of health. The concerns regarding the inclusion of race in GFR estimating equations have prompted discussions and actions aimed at addressing this issue. One significant development has been the removal of the race coefficient from some GFR estimating equations.⁷

In 2021, NICE recommended estimating kidney function using the CKD-EPI equation without ethnicity adjustments. Although the CKD-EPI 2021 equation without ethnicity adjustment has been introduced, it has only been validated in North America. As a result, NICE still recommends using the CKD-EPI 2009 equation.⁸ The eGFR estimation by the new equation consistently showed a lower prevalence of CKD and higher GFR estimates in Asian cohorts compared to the CKD EPI 2009 equation.⁹ The 2009 CKD-EPI creatinine equation still performs well within recommended targets for estimating GFR in the non-Black race group, as measured by the percentage of eGFRs within 30% of measured GFR (mGFR), also known as P30.⁸ The KDIGO 2024 CKD CPG recommends the use of the CKD-EPI equation within geographical regions without race correction but does not specify whether the 2009 or 2021 equation should be used.⁶

The laboratory plays a pivotal role in implementing the standards of care to ensure accuracy and reliability when measuring GFR using creatinine. To ensure accurate measurement of filtration markers, the laboratory should utilise a specific and precise assay with stringent performance criteria. Hence, the KDIGO 2024 CPG advises the laboratory to use an assay with a coefficient of variation (CV) less than 2.3% for creatinine and that the selected assay demonstrates desirable bias less than 3.7% compared with the reference methodology.⁶ Meeting the suggested precision and bias targets ensures that laboratory errors have minimal impact, resulting in less than a 10% increase in root mean square error when estimating GFR.⁶ The recommendation emphasises the preference for utilising enzymatic methods, whenever feasible, for the measurement of creatinine. Enzymatic assays, while not entirely immune to interferences, are generally more specific for creatinine and less vulnerable to chemical and chromogenic interferences like icterus and haemolysis, thus reducing the likelihood of interference in the majority of cases.¹⁰

2. Urine albumin

In the KDIGO 2012 CPG, the terminology used was albuminuria rather than proteinuria as albumin is the principal component of urinary protein in most kidney diseases.² The MACB-CKD Task Force also recommended the use of the term albuminuria rather than microalbuminuria.⁴ The KDIGO 2024 CPG recommends avoiding the use of urine reagent strips unless they can specifically measure urine albumin at low concentrations and express the result as urine albumin-to-creatinine ratio (uACR) for identifying proteinuria in adults. The descending order of preference for testing of albuminuria is (i) uACR, or (ii) reagent strip urinalysis for albumin and ACR with automated reading. A first-morning midstream urine sample is preferred in all cases. When measuring urine protein, the order of preference is: (i) urine protein-to-creatinine ratio (uPCR), (ii) reagent strip urinalysis for total protein with automated reading, or (iii) reagent strip urinalysis for total protein with manual reading. Urine albumin measurement is a more sensitive and specific indicator of alterations in glomerular permeability compared to total urinary protein levels.¹¹ There are factors that can affect the urine albumin and creatinine measurements. When a random untimed urine shows albuminuria, a repeat uACR on a first void morning urine shall be done to confirm it. When uACR is 70 mg/mmol or more, which is a sign of nephrotic syndrome, uPCR can be an alternative measure of proteinuria.¹²

It has also been recommended to implement handling standards to ensure the accuracy and reliability of findings when assessing urine samples.⁶ Adoption of the reporting and handling standards for the assessment of urine samples is of paramount importance to ensure accurate diagnoses, appropriate medical interventions, and ultimately, better patient outcomes. To standardise the reporting format, it was recommended to report uACR as mg/mmol or mg/g and results should be expressed to one decimal place (mg/mmol) or whole numbers (mg/g). The target analytical variation (CV_a) should ideally be less than 0.25 of the biological variation (CV_b), which is approximately 15%.¹³

The new KDIGO guideline also mentions that point-of-care testing (POCT) for creatinine and urine albumin measurements can be utilised in settings where access to a laboratory is limited.⁶ When employing POCT, it is advised to adhere to the same pre-analytical, analytical, and post-

analytical quality criteria regarding specimen collection, device performance, external quality assessment, and result interpretation as for laboratory-based testing.

3. Kidney Failure Risk Equation

The CKD staging heatmaps illustrate relative risks for each CKD category compared to individuals without CKD at a population level. However, determining a person's absolute risk for each outcome necessitates utilising risk prediction equations tailored to the specific adverse event. Individual-level risk prediction can significantly enhance key clinical decisions, foster better patient-healthcare provider communication, and facilitate personalised care for individuals with CKD.¹⁴ For individuals with CKD stages G3–G5, it is recommended to utilise an externally validated risk equation to estimate the absolute risk of kidney failure.⁶ It is important to note that risk prediction equations developed for individuals with CKD stages G3–G5 are not valid for use in those with CKD stages G1–G2. The recommendation also emphasises the importance of externally validated prediction equations that are adaptable to diverse healthcare settings. Furthermore, it highlights the need for implementing laboratory information systems and electronic medical records to facilitate the delivery of risk-based care for individuals with CKD. This is a strong recommendation, as the Working Group has determined that the evidence supporting both the clinical validity and clinical utility of risk prediction equations was robust enough to warrant widespread adoption. The three validated models are: The Kidney Failure Risk Equation (KFRE)¹⁵, the Veterans Affairs model¹⁶, and the Z6 Score model.¹⁷

KFRE was developed and initially validated in 8,391 adults from two Canadian provinces, and subsequently validated in 721,357 individuals from over 30 countries across four continents.¹⁸ In this extensive validation study, cohorts from both general populations and nephrology clinic settings were included, demonstrating excellent discrimination with a C-statistic more than 0.80 in 28 out of 30 cohorts. Four kidney failure risk equations were developed in the original cohorts: the 3-variable equation (age, sex, and eGFR), the 4-variable equation (3-variable plus uACR), the 6-variable equation (4-variable plus diabetes mellitus and hypertension), and the 8-variable equation (4-variable plus calcium, phosphate, bicarbonate, and albumin). Among these, the 4-variable and 8-variable equations showed the

best performance in the original cohorts.¹⁹ The 4-variable equation requires age, sex, eGFR, and uACR. This concise set of variables makes it practical and straightforward, enabling automated risk reporting by the laboratory whenever laboratory tests are performed. The NICE CKD guidelines also recommend using the UK four-variable KFRE.⁶ Hence, in line with NICE CKD guidelines, integrating 4-variable KFRE in the laboratory report helps clinicians in making informed decisions about patient management. KFRE can be reported as a 5-year risk and a 2-year risk. A 5-year kidney failure risk assessment can complement the criteria based on eGFR or uACR and other clinical considerations to determine the need for nephrology referral.¹⁴ A 2-year kidney failure risk can inform the timing of multidisciplinary care in addition to eGFR-based criteria.¹⁴

Based on KDIGO 2024 guidelines, the MACB-CKD Task Force has updated their recommendations to be adopted by the clinical laboratory as follows:

Recommendations for laboratory reporting of eGFR:

Pre-analytical

1. Separate serum/plasma from red blood cells by centrifugation within 12 hours of venepuncture.

Analytical

2. Enzymatic method for creatinine assay shall be used, where possible.
3. Creatinine measurement shall be traceable to isotope-dilution mass spectrometry (IDMS).
4. The analytical imprecision of the creatinine assay should be less than 2.3% with a bias of less than 3.7% compared with reference methodology [or appropriate international standard reference method group target in external quality assessment (EQA)].

Post-analytical

5. Serum creatinine concentration shall be rounded to the nearest whole number when expressed in $\mu\text{mol/L}$.
6. The CKD-EPI 2009 equation shall be used in the calculation of eGFR without using the race correction factor.
7. eGFR shall be rounded to the nearest whole number and reported using the unit mL/min/1.73 m^2 .
8. Clinical laboratories shall report eGFR as $>90 \text{ mL/min/1.73 m}^2$ if it is more than $90 \text{ mL/min/1.73 m}^2$.

9. eGFR levels less than 60 mL/min/1.73 m² shall be flagged as being low.
10. eGFR has to be interpreted with caution in people with extremes of muscle mass or certain illnesses (e.g., bodybuilders, people who have had amputations, muscle wasting disorders, acute myocardial infarction and acute kidney injury).
11. Risk prediction equations developed for use in individuals with CKD G3–G5, are not valid for use in those with CKD G1–G2. In people with CKD G3–G5, the laboratory shall use KFRE risk equation (4 variables) to estimate the absolute risk of kidney failure and shall include
 - a. A 5-year kidney failure risk (low risk <5% and high risk >15%) and
 - b. A 2-year kidney failure risk (low risk <10% and high risk >20%)

Recommendations for laboratory reporting of urine albumin:

Pre-analytical

1. The first void urine in the morning shall be used for the measurement of uACR.
2. Samples for urine albumin measurement shall not be stored at -20 °C.
3. Samples for urine albumin measurement shall be analysed fresh or stored at 4 °C for up to 7 days.

Analytical

4. Urine reagent strips can only be used if they are capable of specifically measuring urine albumin at low concentrations and expressing the result as albumin:creatinine ratio (uACR).
5. Analytical CV of methods to measure urine albumin shall be less than 15%.

Post-analytical

6. For the initial detection of proteinuria:
 - a. use uACR rather than uPCR because of the greater sensitivity for low levels of proteinuria.
 - b. repeat uACR between 3 mg/mmol and 70 mg/mmol in a subsequent early morning sample to confirm the result.
 - c. A repeat sample is not needed if the initial uACR is 70 mg/mmol or more.
7. For quantification and monitoring of large amounts of urine protein (more than 70 mg/mmol), uPCR may be used as an alternative.
8. If unexplained proteinuria is an incidental finding on a reagent strip, testing for CKD using eGFR and uACR is recommended.

9. uACR shall be reported to 1 decimal place whether in mg/mmol or mg/g.

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